CUSTOMER CHURN ANALYSIS

1. Business Problem

The telco dataset concerns telecommunications services, such as phone service, internet service, and technical support. Looking at the dataset, the following questions come to mind;

1. What percentage of customers have churned?
2. What contract type churns the most?
3. Does tenure affect churn?
4. Are customers paying high monthly charges most likely to churn?
5. Are senior citizens a significant percentage of churners?

2. Data

Item	Detail
Source	Telco Customer Churn (Kaggle)
Period	Quarter 3
Size (after cleaning)	7043 customers, 1,869 churned

Key columns

Column	Description
Senior citizen	Indicates if the customer is 65 or older: Yes, No
Tenure	Indicates the total number of months that the customer has been with the company by the end of the quarter specified above.
Phone service	Indicates if the customer subscribes to home phone service with the company: Yes, No

Internet service	Indicates if the customer subscribes to Internet service with the company: No, DSL, Fiber Optic, Cable.
Tech support	Indicates if the customer subscribes to an additional technical support plan from the company with reduced wait times: Yes, No
Contract	Indicates the customer's current contract type: Month-to-Month, One Year, Two Year.
Monthly Charges	Indicates the customer's current total monthly charge for all their services from the company.
Total charges	Indicates the customer's total charges, calculated to the end of the quarter specified above.
Churn	1 = the customer left the company this quarter. 0 = the customer remained with the company. Directly related to Churn Label.

3. Data Cleaning

Tool: Excel

Step	What I did	Result
Duplicate records	Checked the dataset for duplicate customer records.	No duplicate records were found.
Missing values	Checked for missing values and identified null values in the TotalCharges column for customers with zero tenure and replaced nulls with 0.	Confirmed that the affected customers had zero tenure and no accumulated total charges.
Data types	Checked that columns containing numerical values, categories, and customer identifiers had appropriate data types.	Data types were checked and corrected where necessary.

4. Key Findings

Tool: SQL [PostgreSQL]

Q1. What is the overall churn rate?

Of 7,043 customers, 1,869 churned, representing a churn rate of 26.54%.

Total Customers	Churned Customers	Churn Rate
7043	1869	26.54

Key Finding: The overall customer churn rate is 26.54%, meaning that approximately one in four customers left the company during the period analyzed. This establishes a baseline for evaluating customer retention and identifying the customer segments contributing most to churn.

Q2. Which contract has the highest churn?

Contract	Total Customers	Churned Customers	Churn Rate
Month-to-month	3875	1655	42.71
One year	1473	166	11.27
Two year	1695	48	2.83

Key Finding: Month-to-month customers have the highest churn rate at 42.71%, compared with 11.27% for one-year contracts and 2.83% for two-year contracts. This indicates a strong association between shorter contract duration and higher customer churn.

Q3. Do monthly charges relate to churn?

Churn Status	Total Customers	Average Monthly Charges
Did not churn	5174	$61.27
Churned	1869	$74.44

Key Finding: Customers who churned paid an average of $74.44 per month, compared with $61.27 among customers who remained. This suggests an association between higher monthly charges and customer churn. Further analysis

is needed to determine whether pricing, service combinations, or other customer characteristics contribute to this difference.

Q4. Does tenure relate to churn?

Tenure Group	Total Customers	Churned Customers	Churn Rate
0-12 months	2186	1037	47.44
13-24 months	1024	294	28.71
25-48 months	1594	325	20.39
49-72 months	2239	213	9.51

Key Finding: Churn rates decrease as customer tenure increases, falling from 47.44% among customers with 0–12 months of tenure to 9.51% among those with 49–72 months. This indicates that newer customers are substantially more likely to churn than longer-standing customers, highlighting the early customer lifecycle as an important area for further investigation.

Q5. Does internet service relate to churn?

Internet Service	Total Customers	Churned Customers	Churn Rate
Fiber optic	3096	1297	41.89
DSL	2421	459	18.96
No	1526	113	7.40

Key Finding: Fiber-optic customers have the highest churn rate at 41.89%, compared with 18.96% for DSL customers and 7.40% for customers without internet service. This reveals a substantial difference in churn across service categories, warranting further investigation into pricing, service quality and customer experience.

Q6. Does technical support relate to churn?

Tech Support	Total Customers	Churned Customers	Churn Rate
No	3473	1446	41.64
Yes	2044	310	15.17

No internet service	1526	113	7.40

Key Finding: Customers without technical support have a churn rate of 41.64%, compared with 15.17% among customers with technical support. This indicates a strong association between technical support subscription status and churn, although further analysis is needed to determine whether support availability itself explains the difference.

Q7. Do senior citizens make up a significant percentage of Churned customers?

Senior Citizen	Total Customers	Churned Customers	Churn Rate
1	1142	476	41.68
0	5901	1393	23.61

$$\%\ of Churners\ that\ are\ senior\ citizens = \frac{476}{1869} \times 100 = 25.46\%$$

Key Finding: Senior citizens have a churn rate of 41.68%, compared with 23.61% among non-senior customers. Although senior citizens account for 25.46% of all churned customers, the higher churn rate suggests they are more likely to leave than their non-senior counterparts in this dataset.

5. Business Recommendation

- Improve first-year retention

  The first-year group has a 47.44% churn rate. Investigate early customer experiences and pilot an onboarding programme. This could include a welcome message, guidance on using subscribed services, an early check-in to identify technical problems, and proactive support for customers experiencing difficulties.

- Address month-to-month churn

  The 42.71% churn rate makes this a strong candidate for targeted retention offers and contract-specific analysis. Review the pricing and service combinations of customers with high monthly charges,

particularly those who are also on month-to-month contracts. Avoid giving every high-paying customer a discount without first identifying the underlying issue.

- Investigate fiber-optic churn

  The 41.89% churn rate warrants a deeper investigation into service experience, pricing and customer segments before deciding on an intervention.

- Test targeted technical support

  Evaluate whether support interventions improve retention enough to justify their cost.

6. **About**

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